

ANM-D5C-0003 DDR5 SDRAM

Synthetic technical datasheet for Anistroph and Aina-Veris testing. Product identifiers are fictional; engineering parameters use realistic memory-industry domains.

Product Summary

Product	Tech	Density	Rate	Width	Temp	Package
ANM-D5C-0003	DDR5	32 Gb	5200 MT/s	x16	-40..95 C	VFBGA

Power Management

DDR5-class products support controller-managed idle, power-down, and self-refresh states. Entry and exit timing must match the configured operating point. Higher transfer rates increase the importance of power-delivery integrity and thermal margin. LPDDR5X products use programmable low-power behavior with additional controller-managed state transitions.

Initialization

After stable power and clock conditions, the controller performs reset release, mode-register configuration, timing setup, and required training before normal traffic. The timing profile must correspond to the selected transfer rate; lower-speed timing assumptions must not be reused at higher operating points.

Temperature and Operating Restrictions

A required operating point is valid when it falls inside the product's stated minimum and maximum temperature range. For example, a 55 C requirement is satisfied by a product rated 0..95 C or -40..95 C. The value 55 C is therefore interpreted as a range-containment requirement rather than a discrete catalog grade.

Revision S1 - synthetic reference document; not a commercial component datasheet.